Majid Farhadi

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Education

University of Alberta, Computer Engineering — Edmonton, Canada, 2023 – 2026

Isfahan University of Technology, Computer Engineering — Isfahan, Iran, 2018 – 2023

Experience

Very Large Scale Integration (VLSI) Research Lab, University of Alberta, Graduate Research Assistant — Edmonton, Canada, 2023 – 2026, 3 years

- Designed and implemented AI architectures, including training and inference circuits, on FPGAs.
- Integrated and synchronized IP core modules with edge detector 1-stage synchronizers.
- Wrote testbench modules for behavioral, functional, and timing simulation and verification of RTL designs.
- Defined implementation constraints including clock period to meet timing requirements.
- Optimized dynamic power by minimizing switching activity through stop conditions.
- Fine-tuned distance functions in clustering algorithms and derived gradients in backpropagation algorithms.

Structured Machine Learning Research Lab, Simon Fraser University, Graduate Research Assistant — Burnaby, Canada, 2023 – 2024, 1 year

- Counted motifs in a graph database by matrix multiplication, suitable for ML applications.

Embedded Systems and Networks Research Lab, Isfahan University of Technology, Undergraduate Research Assistant — Isfahan, Iran, 2021 – 2023, 2 years

- Designed an interface for a hardware accelerator for the time-slotted channel hopping (TSCH) protocol.
- Stacked on top of an IEEE 802.15.4 transceiver used in wireless sensor networks in IoT applications.
- Deployed on an Altera DE1-SoC board, featuring a Cyclone FPGA and a dual-core Cortex-A9 ARM.
- Integrated the ARM processors and FPGA fabric through a lightweight master-slave AXI bridge.
- Enabled full-duplex read/write communication between FPGA and ARM processor.
- Compiled C programs for the target ARM architecture using makefiles.
- Accessed the FPGA from C programs via virtual addresses provided by memory mappings.
- Offloaded computation from the processor and exploited the FPGA's high throughput.

Kian-Pardaz Naghsh-e-Jahan Engineering Company, Intern — Isfahan, Iran, 2021 – 2021, 1 year

- Designed a PCB and implemented a web client for a smart thermostat on an ESP8266-12E module.

Publications

SCFNNs: Stochastic Computing Fuzzy Neural Networks
Majid Farhadi, Witold Pedrycz, Jie Han
ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11360989

Rule-Enhanced Graph Learning
Ali Khazraee, Abdolreza Mirzaei, Majid Farhadi, Parmis Nadaf, Kiarash Zahirnia, Mohammad Salameh, Kevin Cannons, Richard Mar, Mingyi Wu, Oliver Schulte
openreview.net/forum?id=m02qzfjlHA&referrer=%5Bthe%20profile%20of%20Kiarash%20Zahirnia%5D(%2Fprofile%3Fid%3D~Kiarash

1BCMs: 1-Bit Cognitive Maps
Majid Farhadi, Witold Pedrycz, Jie Han

A Memory-Efficient and Low-Complexity Probabilistic Sigmoidal Approximator
Wenhao Lu, Majid Farhadi, Xingjian Gao, Chris Lynch, Jie Han

Enhancing Graph Generation With First-Order Logic Rules
Ali Khazraee, Abdolreza Mirzaei, Majid Farhadi, Parmis Nadaf, Kiarash Zahirnia, Mohammad Salameh, Kevin Cannons, Richard Mar, Mingyi Wu, Oliver Schulte

Skills

Programming Languages

Tools and Frameworks

Languages

English
Fluent

Persian
Native speaker

Interests

Stochastic Computing

Fuzzy Sets

Neural Networks

FPGAs

Hardware Acceleration

Computer Architecture

Computer Vision

Cryptography

Projects

Soft-Core Implementation of MIPS Architecture on FPGA

Adding System Call to XV6 Kernel

Development of Smart Health Bracelet to Communicate Vital Signs by Arduino Board and GSM Module

ECG Signals Analysis on IoT Platform by Autoencoders

Prediction of DDoS Botnet Attacks on IoT Devices by Data Mining Methods

Implementation of Autonomous Obstacle Avoiding Robot by Arduino Board and Ultrasonic Sensor

Image Seam Carving based on Depth Map

Time-Series Forecasting with Deep and Recurrent Neural Networks

References

Dr. Jie Han

Dr. Witold Pedrycz

Dr. Oliver Schulte